Significant figures, precision and accuracy

In this sheet you will…

- count significant figures and round to them
- tell the difference between accuracy and precision
- decide how many digits a measurement deserves
- see why rounding cannot make a measurement more accurate

Key words

significant figures
accuracy
precision
true value
spread · mean
false precision

This sheet follows our lesson step by step, so you can read it again at home.

1 Why we need significant figures

Last lesson we rounded to **decimal places**. In the starter, a scientist measured a cell: **0.0048 mm**.

0.0048 mm (a cell) → to 2 decimal places: **0.00**. Is that useful? **No, the measurement disappeared.**

Decimal places count from the decimal point, so they ignore whether a number is big or tiny. We need a way of counting that starts where the information starts.

Significant figures (s.f.)

Start counting at the **first non-zero digit**. Zeros in front are only placeholders.

Example: 0.00472

The zeros at the front do not count. The first digit that matters is the 4, so the digits are 4 7 2: **3 s.f.**

Three rules for counting

1. Start at the first non-zero digit.
2. Zeros **between** digits count.
3. Zeros at the **end** count if they are after the decimal point.

Worked example 1: how many significant figures?

Number	Digits that count	s.f.	Why
5003	5 0 0 3	4	zeros between digits count
0.0480	4 8 0	3	the last zero is after the decimal point, so it was measured
70.0	7 0 0	3	the two zeros after the 7 count

Careful: 5000 on its own is ambiguous. It could have 1, 2, 3 or 4 significant figures.

2 Rounding to significant figures

Same rounding rule as last lesson: find the last digit you keep, look at the next one, and round up if it is 5 or more. Only the **counting** is new.

Worked example 2

a **4.5678 to 3 s.f.** Digits 4, 5, 6. The next digit is 7, so round up: **4.57**

b **0.04328 to 3 s.f.** The first non-zero digit is the 4: 4, 3, 2. The next digit is 8, so round up: **0.0433**

c **45 678 to 2 s.f.** Digits 4, 5. The next digit is 6, so round up: 46. But the answer is not 46: the size must stay. **46 000**

d **45 678 to 3 s.f.** Digits 4, 5, 6. The next digit is 7: **45 700**

Decimal places or significant figures? Which one keeps the measurement?

Number	to 2 d.p.	to 2 s.f.
0.0048	0.00	0.0048
0.00472	0.00	0.0047

Decimal places **fix a position**. Significant figures **keep a number of meaningful digits**.

Did you know? 0.0048 mm is the same length as 4.8 micrometres. The numbers are different, but both have 2 significant figures. Decimal places change with the unit (4 d.p. and 1 d.p.), significant figures do not.

Exercise 1 significant figures

1 **a** How many significant figures? 0.00472 5003 0.0480 70.0 2.50 0.0905 500.56
b Is 1200 clear? Explain why not.

2 Round to **3 s.f.**: a) 4.5678 b) 0.04328 c) 0.8996 d) 1.0049 e) 45 678

3 Round to **2 s.f.**: a) 45 678 b) 0.07351 c) 3.049 d) 0.0605 e) 12.96

4 Round each number to 2 decimal places and to 2 significant figures.

Number	2 d.p.	2 s.f.
12.3456		
0.0605		
0.00093		

5 A cell is 0.0036 mm long, which is 3.6 micrometres. Give the number of decimal places and of significant figures of each form.

6 Tom rounds 0.03456 to 2 s.f. and writes 0.03. What did he do wrong? Write the correct answer.

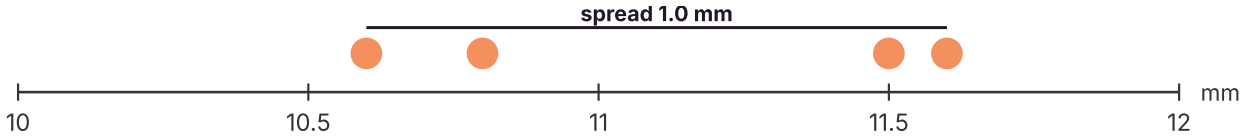
7 Round 6.0396 to 3 s.f. and to 4 s.f. Why does the second answer keep a zero at the end?

8 Write a number between 0 and 1 that has exactly 2 significant figures and 4 decimal places.

Precision and accuracy

3 Precision and accuracy: the bread slice

Someone on the internet (**@germanbreadcutter**) cut a slice of bread, saying *"let's see how accurate we can cut today"*. He measured how thick the slice is at its four sides, in millimetres.

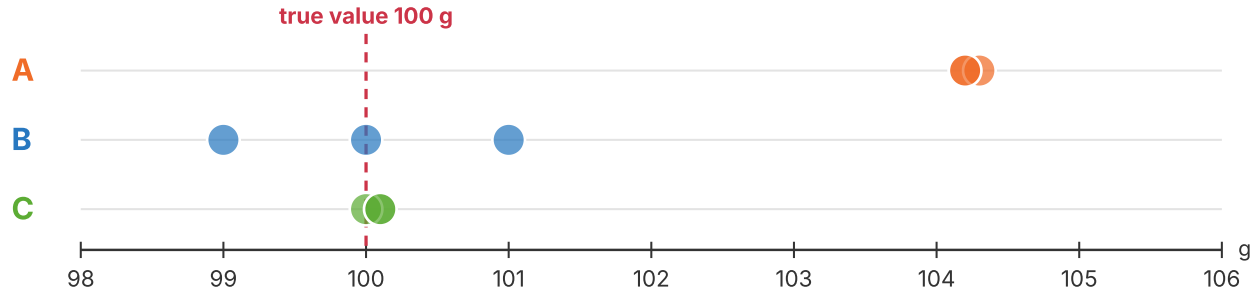


11.6 10.8 10.6 11.5 → spread = thickest − thinnest = 11.6 − 10.6 = 1.0 mm

The four sides are not equally thick. A bread-cutting machine doing the same job has a spread of 0.70 mm, so the machine cuts **more evenly**: a smaller spread means more consistent.

A different experiment: three scales, one true mass

A mass of exactly **100 g** is weighed three times on each scale.



A 104.2, 104.3, 104.2: the readings agree, but all are about 4 g too high.

B 99, 101, 100: the readings are spread out, but the average is 100 g.

C 100.1, 100.0, 100.1: the readings agree and are close to 100 g.

So we need two different words:

Accuracy

How close a measurement is to the **true value**.

The question is: **is it right?**

Precision

How close repeated measurements are to **each other**.

The question is: **is it consistent?**

Scale	Precise?	Accurate?	Verdict
A	yes	no	precise, not accurate
B	no	yes, on average	accurate, not precise
C	yes	yes	both

To check accuracy you need the true value (100 g). To check precision you only need the readings. The baker measured **precision**: we did not need the true thickness to find his spread. For accuracy we would need a target: if he wanted 10 mm, an average of about 11 mm is too thick.

Precision and accuracy

4 How many digits should he write?

Let's find the average thickness of the slice. Add the four readings and divide by 4:

11.6 + 10.8 + 10.6 + 11.5 = 44.5 → mean = 44.5 ÷ 4 = 11.125 mm

Should he write 11.125 mm? That many digits claims that we know the thickness to a **thousandth of a millimetre**. But the sides of the slice differ by a whole millimetre. His tool shows hundredths and he writes only one decimal: the honest choice, because the hundredths carry no information.

11.125 → 11.1 mm (11.125 has 5 s.f. · 11.1 has 3 s.f.)

Rounding to fewer significant figures makes a **more modest claim** about precision. That is why scientists use significant figures: the digits we write are a promise about precision.

Watch out!

More digits does not mean more accurate. Writing more digits than the data supports is **false precision**. **Precise does not mean correct.** Scale A gives the same number every time, and is wrong every time.

5 Does rounding make it accurate?

Scale A says 104.2 g for the 100 g mass. We round it to 104 g. Is it accurate now? **No**, it is still 4 g too high.

Rounding changes
how many digits we write, so the **precision we claim**.

Rounding never changes
how close the measurement is to the true value: its **accuracy**.

Summary: three lines for the board

Accuracy = right. **Precision** = consistent. **Digits** = a promise about precision.

Exercise 2 precision and accuracy

1 A mass of exactly **50 g** is weighed three times on each of four scales.

Scale	Reading 1	Reading 2	Reading 3
P	50.1 g	49.9 g	50.0 g
Q	53.4 g	53.5 g	53.4 g
R	48 g	52 g	50 g
S	45 g	56 g	58 g

a For each scale, write accurate, precise, both or neither.
b Which scale would you buy? Why?
c Round the readings of scale Q to the nearest gram. Are they now accurate? Explain.

> **Tip** Accuracy: look at the mean of the readings. Precision: look at how far apart the readings are.

2 Maria cuts three slices of cheese. She wants each to be **5.0 mm** thick. She measures 4.1 mm, 4.0 mm, 4.2 mm.
a Are her slices precise? **b** Are they accurate? Explain both. **c** What advice would you give her?

3 A tool shows hundredths of a centimetre. Jon measures a pencil four times: 14.52 cm, 14.47 cm, 14.58 cm, 14.51 cm.
a Work out the mean. **b** Find the spread (largest − smallest).
c The hundredths change from reading to reading, so give the mean to 1 decimal place. Write it, and say how many significant figures it has.

4 A ruler marked only in whole centimetres gives the length of a block as about 7.4 cm. Rick divides it by 3. His calculator shows 2.4666666... He writes 2.5 cm.
a Why is it a bad idea to write all the digits? **b** Does rounding to 2.5 cm make the answer more accurate? Explain.

5 Scale B reads 99 g, 101 g, 100 g for a 100 g mass. Give the mean. A student writes “mean = 100.000 g”. What is wrong with that?

CHALLENGE
Invent three readings of a 20 g mass that are precise but not accurate. Then invent three that are accurate on average but not precise.

What do you think?

A scale shows **1.00000000 kg** for a mass.
1 Is the scale accurate? Is it precise? How can you tell?
2 What does the number of digits tell you? What does it not tell you?

Summary checklist

- ☐ I can count significant figures and round to them.
- ☐ I can explain accuracy and precision in my own words.
- ☐ I can decide if readings are accurate, precise, both or neither.
- ☐ I know the digits I write are a promise about precision.
- ☐ I know that rounding cannot make a measurement more accurate.